

EXP 3

CLASSIFICATION WITH DECISION TREE

AIM:

To implement a Decision Tree classifier and evaluate its performance using accuracy score and

confusion matrix on a real-world dataset.

Import Libraries:

```
[20]: from sklearn.datasets import load_iris
      from sklearn.tree import DecisionTreeClassifier, plot_tree
      from sklearn.model_selection import train_test_split
      from sklearn.metrics import confusion_matrix, accuracy_score
      import matplotlib.pyplot as plt
      import seaborn as sns
```

Load Dataset & Train-Test Split

```
[2]: iris = load_iris()
     X = iris.data
     y = iris.target
```

```
•[13]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

Train the Decision Tree

```
[21]: dt_model = DecisionTreeClassifier(criterion='gini', random_state=0)
      dt_model.fit(X_train, y_train)
```

```
[21]: ▾ DecisionTreeClassifier ⓘ ?
      ▶ Parameters
      ▶ Fitted attributes
```

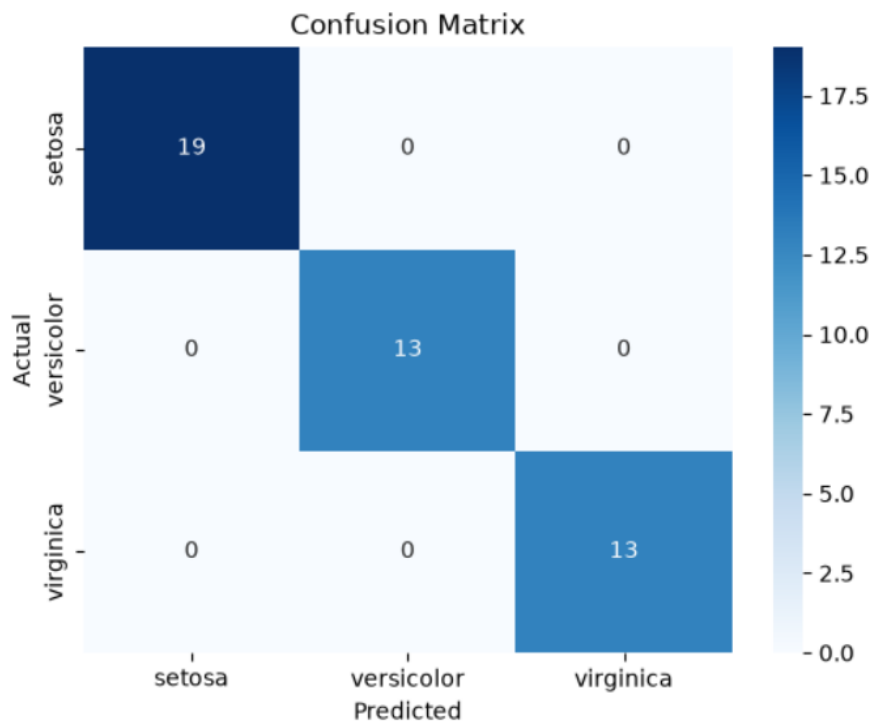
Prediction & Accuracy Output

```
[23]: cm = confusion_matrix(y_test, y_pred)
      acc = accuracy_score(y_test, y_pred)
      print("Confusion Matrix:\n", cm)
      print("Accuracy Score:", acc)
```

```
Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
Accuracy Score: 1.0
```

Confusion Matrix Heatmap

```
[24]: sns.heatmap(cm, annot=True, cmap="Blues", xticklabels=iris.target_names,
      yticklabels=iris.target_names)
      plt.xlabel("Predicted")
      plt.ylabel("Actual")
      plt.title("Confusion Matrix")
      plt.show()
```

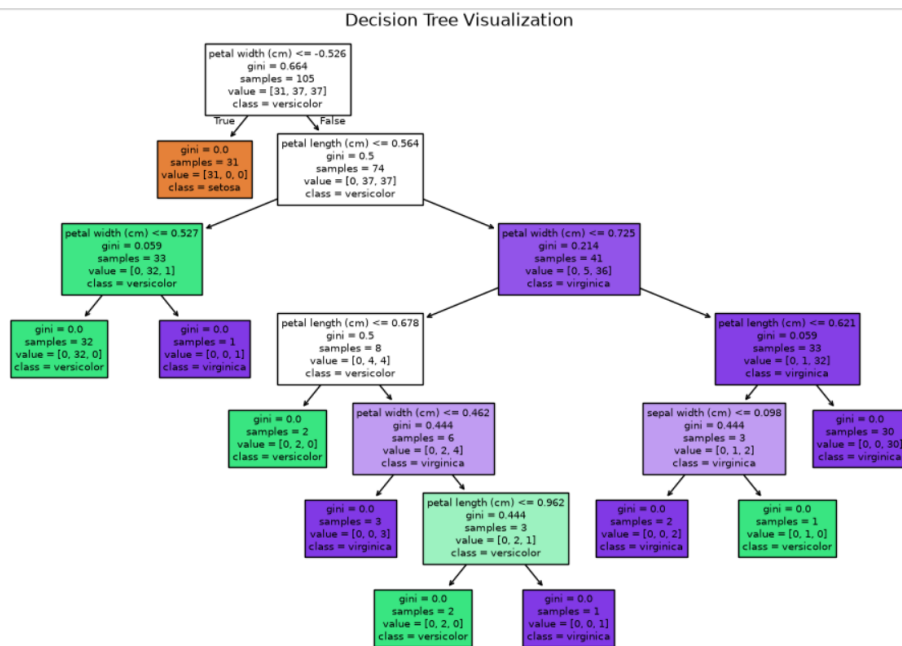


Decision Tree Visualization

```
[17]: rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
      rf_model.fit(X_train, y_train)
```

```
[17]: RandomForestClassifier ⓘ ⓘ
      Parameters
      Fitted attributes
```

```
[25]: plt.figure(figsize=(12,8))
      plot_tree(dt_model, filled=True, feature_names=iris.feature_names, class_names=iris.target_names)
      plt.title("Decision Tree Visualization")
```



RESULT:

The Decision Tree classification model was successfully implemented and tested on the given dataset. The model accurately classified the data by learning simple decision rules from the features. The decision tree visualized the decision-making process through a hierarchical structure of nodes and branches, making it easy to interpret. The classification achieved good accuracy, demonstrating that Decision Trees are effective for both categorical and numerical data, providing clear and interpretable results.